



QP CODE: 23709212



Reg No :

Name :

M.C.A. DEGREE EXAMINATION, MAY 2023

First Semester

Master of Computer Application

Core - MCACT102 - "DIGITAL LOGIC & COMPUTER ORGANIZATION"

2020 Admission Onwards

2F6CD684

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. What are the steps for converting Gray code to binary and vice versa? Give examples.
2. What is meant by parity bit?
3. Write a note on alphanumeric codes.
4. State the basic laws of Boolean Algebra.
5. What are SOP and POS forms of Boolean expression? Give example.
6. Define Sequential circuit.
7. What are the basic functional units of a computer?
8. What is RISC?
9. Distinguish between RAM and ROM.
10. Explain Program controlled I/O.
11. Explain the need for interconnection networks in multiprocessor systems.
12. What are the hazards in pipelining?

(10×3=30 marks)





Part B

Answer *all* questions

Each question carries **9** marks

13. a) Perform subtraction using 2's complement method on the binary and explain the steps.

i) $1011011011 - 101101$

ii) $10101011 - 101010$

iii) $11.10 - 110.1$

OR

- b) Explain how to detect and correct errors in the following odd parity Hamming code 0101101

14. a) Solve the following using K-Map. $F(A,B,C, D) = \sum(1,3,9,11,4, 5, 12, 13, 10,14)$.

OR

- b) Explain the types of flipflops.

15. a) Describe the various addressing modes.

OR

- b) Explain the micro-programmed control unit with a diagram.

16. a) Explain Random Access Memory in detail.

OR

- b) Explain Direct Memory Access with suitable diagrams.

17. a) Explain the Flynn's classification architecture of parallel processing.

OR

- b) Explain pipelining, types of pipelining and hazards of pipelining.

(5×9=45 marks)

